

3.5 Control Structures

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

CS XI

Chapter 3: Programming Fundamentals

Lecture 13 of 24

★ Golden

3.5 Control Structures ★ Golden

LEARN IT

Control structures decide the **order in which statements are executed** in a program (the "flow of control"). Almost every program is built from just three basic structures:

1. **Sequence** — do the steps one after another, top to bottom.
2. **Selection** — choose a path depending on a condition (`if` , `else`).
3. **Repetition** — repeat steps (loops: `for` , `while`).

WORKED EXAMPLE

Morning routine as control structures:

- Sequence: wake up → brush teeth → have breakfast.
- Selection: **if** it is raining, take an umbrella; **else** take a cap.
- Repetition: **while** the bus has not come, keep waiting.

REMEMBER IT

Remember "**S-S-R**": **S**election, **S**election, **R**epetition — "Step, Select, Repeat".

COMMON MISTAKE

Selection and repetition both use conditions, but they are different: selection runs a block **at most once**; repetition can run a block **many times**.

اردو میں سمجھیں

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔ تین بنیادی اسٹرکچر ہیں: سیکوئنس (ترتیب وار)، سلیکشن (شرط کی بنیاد پر انتخاب) اور ریپیٹیشن (دہرانا یعنی لوپ)۔

EXAM TIP

"Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

CHECK YOURSELF

1. Name the three basic control structures.
2. Which structure is used for "keep asking for the password until it is correct"?

Answers: 1. Sequence, selection, repetition. 2. Repetition (a `while` loop).

Next lecture: 3.5.1 Sequence